USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

Name: Yoon Lee Email: yoonlee@apple.com

Company: Apple Mailstop: _____

Address: _____

City: _____ State/Province: _____

Country: _____ Zip/Postal Code: _____

Phone: _____ FAX: _____

Name: _____ Email: _____

Company: _____ Mailstop: _____

Address: _____

City: _____ State/Province: _____

Country: _____ Zip/Postal Code: _____

Phone: _____ FAX: _____

Name: _____ Email: _____

Company: _____ Mailstop: _____

Address: _____

City: _____ State/Province: _____

Country: _____ Zip/Postal Code: _____

Phone: _____ FAX: _____

Name: _____ Email: _____

Company: _____ Mailstop: _____

Address: _____

City: _____ State/Province: _____

Country: _____ Zip/Postal Code: _____

Phone: _____ FAX: _____

USB PD CTS ENGINEERING CHANGE NOTICE

Title: TEST.PD.PS.SRC.3
Applied to: USB PD CTS Q3, 2026

Brief description of the functional changes proposed:
Update the test post PR Swap to sink per the latest sink transition test(TEST.PD.PS.SNK.1) in Q3 release(https://groups.usb.org/wg/pdcompli/document/18837

Benefits as a result of the proposed changes:
DUTs in Sink role tested per the base spec timing definition.

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
None

An analysis of the software implications:
None

An analysis of the compliance testing implications:
Tester to update the corresponding tests per the ECR updates.

An analysis of the Vendor Info File (VIF) implications:
None

USB PD CTS ENGINEERING CHANGE NOTICE

Actual Change Requested

TEST.PD.PS.SRC.3

From Text:

TEST.PD.PS.SRC.3 Initial Source PDO Transition Post PR Swap

Description: The Tester verifies that after a *PR_Swap*, the new Sink UUT responds correctly to a new *Source_Capabilities* Messages

Test Specific Tester Behavior:

- The Tester is a Dual-Role Power, initially a Sink.
- The Tester sends *Sink_Capabilities* setting B27 (Unconstrained Power) to 1b COMMON.PROC.PD.3

Test Conditions:

	Consumer Only, C/P	Provider Only, P/C, DRP
Rev2Src		✓
Rev3ChkdSrc		✓

Test Procedures:

1. The Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. After this point the tester should reject any *PR_Swap* request.
3. The Tester sends a *PR_Swap* Message.
4. The Tester checks the response. [TEST.PD.PS.SRC.3#1]
    a. If the UUT responds with a *Wait* Message and the Tester resends *PR_Swap* every *tPRSwapWait* + **Error! Reference source not found.** for ***Error! Reference source not found.*** times. Once the Tester has resent *PR_Swap* for ***Error! Reference source not found.*** times, the check fails, and the test ends here.
    b. If the UUT responds with *Reject* or *Not_Supported* checkVIF field PD_Port_Type is set to "Provider Only" or VIF field Accepts_PR_Swap_As_Src = No. The test ends here.
    c. If the UUT responds with an *Accept* Message, check that the VIF field PD_Port_Type is not set to "Provider Only" and VIF field Accepts_PR_Swap_As_Src = Yes.
5. After this point the Tester shall *Reject* any *PR_Swap* request.
6. The Tester draws no more than *iSwapStandby* within *tSrcTransition* min after it has sent the last *GoodCRC*.
7. The Tester checks that the UUT sends a *PS_RDY* Message after the UUT driving V_{BUS} voltage to *vSafe0V*. [TEST.PD.PS.SRC.3#2]
8. The Tester checks that the UUT sends *PS_RDY* within *tSrcTransition* max plus *tSrcSwapStdby* max after the last bit of *GoodCRC* was sent in response to the received *Accept*. [TEST.PD.PS.SRC.3#3]
9. The Tester presents *vSafe5V* on V_{BUS} and then it sends a *PS_RDY* Message, both within *tNewSRC* max of the last *GoodCRC* sent by the UUT.
10. The Tester checks that the UUT draws *iSnkStby* within *tNewSnk*. [TEST.PD.PS.SRC.3#19]
11. The Tester transmits a *Source_Capabilities* Message with the Capabilities as defined in COMMON.PROC.PD.2 Paragraph 2c. (as these will have been the *Source_Capabilities* returned if the UUT has requested them during the *PR_Swap*. The Tester checks that the UUT sends a *Request* Message. [TEST.PD.PS.SRC.3#4]
12. The UUT is now a Sink, and the Tester is a Source.
13. The Tester sends an *Accept* Message.
    a. If a new PDO has been requested and the transition involves increasing the voltage:
        i. After a total of *tSrcTransition* min, check that current drawn by the UUT does not exceed *iSnkStdby* max. [TEST.PD.PS.SRC.3#5]

ii. Wait *tSrcTransition* from the *GoodCRC* Message and then change V_{BUS} voltage requested to the new value and/or be prepared to supply more or less current within *tSrcTransition* max plus *tSrcReady* max as appropriate.
iii. Check that the average current drawn by the UUT, measured in the 1ms before sending *PS_RDY*, does not exceed *iSnkStdby* max. [TEST.PD.PS.SRC.3#6]

b. If a new PDO with no change in the voltage or no new PDO requested,
After *tSrcTransition* max + *tSrcReady* max (315ms) time period check that the current drawn from V_{BUS} does not exceed the previously contracted current. [TEST.PD.PS.SRC.3#7]

14. The Tester sends a *PS_RDY* Message to the UUT.
15. The Tester performs the following check [TEST.PD.PS.SRC.3#8]
    a. For Fixed, Battery or Variable PDO:
    The Tester checks that the current drawn by the UUT does not exceed the level specified in the current PDO over the next 5 seconds.
    b. For PPS PDO:
    N/A
16. The Tester presents *SinkTxOk* if the test is in PD3 mode. The Tester waits **Error! Reference source not found.** to respond to messages from the UUT.
17. The Tester sends a new *Source_Capabilities* Message with a single PDO.
    The PDO:
        i. B31…30 (Fixed Supply) set to 00b.
        ii. B29 (Dual-Role Power) set to 1b.
        iii. B28 (USB Suspend Supported) set to 0b.
        iv. B26 (USB Communications Capable) set to 0b.
        v. B25 (Dual-Role Data) set to 1b.
        vi. B24…22 (Reserved) set to 00b.
        vii. B21…20 (Peak Current) set to 00b.
        viii. B19…10 (Voltage) set to 5V.
        ix. B9…0 (Maximum Current) set to the full requested current as provided in the 1st PDO of the *Sink_Capabilities* from the UUT
18. The check fails if the UUT does not responds with a *Request* Message. [TEST.PD.PS.SRC.3#9]
19. The Tester sends an *Accept* Message.
    a. After *tSrcTransition* max + *tSrcReady* max (315ms) time period check that the current drawn from V_{BUS} does not exceed the contracted current. [TEST.PD.PS.SRC.3#12]
    b. The Tester sends a *PS_RDY* Message to the UUT.
20. The Tester checks that the current drawn by the UUT does not exceed the level specified in the current PDO over the next 5 seconds. [TEST.PD.PS.SRC.3#13]
21. The Tester sends a new *Source_Capabilities* Message with a single PDO.
    The PDO:
        i. B31…30 (Fixed Supply) set to 00b.
        ii. B29 (Dual-Role Power) set to 1b.
        iii. B28 (USB Suspend Supported) set to 0b.
        iv. B26 (USB Communications Capable) set to 0b.
        v. B25 (Dual-Role Data) set to 1b.
        vi. B24…22 (Reserved) set to 00b.
        vii. B21…20 (Peak Current) set to 00b.
        viii. B19…10 (Voltage) set to 5V.
        ix. B9…0 (Maximum Current) set to 0mA.
22. The Tester checks that it receives a *Request* Message [TEST.PD.PS.SRC.3#14]
23. The Tester sends an *Accept* Message.
24. After a total of *tSrcTransition* min, check that current drawn by the UUT does not exceed *pSnkSusp* max. [TEST.PD.PS.SRC.3#17]
25. The Tester sends a *PS_RDY* Message to the UUT.

26. The Tester checks that the current drawn by the UUT does not exceed the level specified in the current PDO over the next 5 seconds. [TEST.PD.PS.SRC.3#18]

To Text:

TEST.PD.PS.SRC.3 Initial Source PDO Transition Post PR Swap

Description: The Tester verifies that after a *PR_Swap*, the new Sink UUT responds correctly to a new *Source_Capabilities* Messages

Test Specific Tester Behavior:

- The Tester is a Dual-Role Power, initially a Sink.
- The Tester sends *Sink_Capabilities* setting B27 (Unconstrained Power) to 1b COMMON.PROC.PD.3

Test Conditions:

	Consumer Only, C/P	Provider Only, P/C, DRP
Rev2Src		✓
Rev3ChkdSrc		✓

Test Procedures:

1. The Tester runs bring-up procedure with the UUT as a Source COMMON.PROC.BU.1.
2. After this point the tester should reject any *PR_Swap* request.
3. The Tester sends a *PR_Swap* Message.
4. The Tester checks the response. [TEST.PD.PS.SRC.3#1]
    a. If the UUT responds with a *Wait* Message and the Tester resends *PR_Swap* every *tPRSwapWait* + **Error! Reference source not found.** for ***Error! Reference source not found.*** times. Once the Tester has resent *PR_Swap* for ***Error! Reference source not found.*** times, the check fails, and the test ends here.
    b. If the UUT responds with *Reject* or *Not_Supported* checkVIF field PD_Port_Type is set to "Provider Only" or VIF field Accepts_PR_Swap_As_Src = No. The test ends here.
    c. If the UUT responds with an *Accept* Message, check that the VIF field PD_Port_Type is not set to "Provider Only" and VIF field Accepts_PR_Swap_As_Src = Yes.
5. After this point the Tester shall *Reject* any *PR_Swap* request.
6. The Tester draws no more than *iSwapStandby* within *tSrcTransition* min after it has sent the last *GoodCRC*.
7. The Tester checks that the UUT sends a *PS_RDY* Message after the UUT driving V_{BUS} voltage to *vSafe0V*. [TEST.PD.PS.SRC.3#2]
8. The Tester checks that the UUT sends *PS_RDY* within *tSrcTransition* max plus *tSrcSwapStdby* max after the last bit of *GoodCRC* was sent in response to the received *Accept*. [TEST.PD.PS.SRC.3#3]
9. The Tester presents *vSafe5V* on V_{BUS} and then it sends a *PS_RDY* Message, both within *tNewSRC* max of the last *GoodCRC* sent by the UUT.
10. The Tester checks that the UUT draws *iSnkStby* within *tNewSnk*. [TEST.PD.PS.SRC.3#19]
11. The Tester transmits a *Source_Capabilities* Message with the Capabilities as defined in COMMON.PROC.PD.2 Paragraph 2c. (as these will have been the *Source_Capabilities* returned if the UUT has requested them during the *PR_Swap*. The Tester checks that the UUT sends a *Request* Message. [TEST.PD.PS.SRC.3#4]
12. The UUT is now a Sink, and the Tester is a Source.
13. The Tester sends an *Accept* Message.
14. If a new PDO has been requested and the transition involves increasing the voltage:
    a. If the transition is to a Fixed Supply PDO, or to an AVS APDO with voltage change greater than *vSmallStep*.
        i. After a total of *tSnkStdby* from the EOP of the *GoodCRC* Message acknowledging the *Accept* Message, check that current drawn by the UUT does not exceed *iSnkStdby* max. [TEST.PD.PS.SRC.3#5]

ii. Wait a nominal *tSrcTransition* from the *GoodCRC* Message and then change V_{BUS} voltage to the new requested value. For Fixed Supply PDO transitions, the AVS APDO transitions, the Tester shall complete the voltage change before *tAvsSrcTransLarge* expires.
iii. As soon as the voltage on V_{BUS} has stopped changing, check that current drawn by the UUT does not exceed *iSnkStdby* max until *PS_RDY* is sent. [TEST.PD.PS.SRC.3#6]

b. If the transition is to an AVS APDO with a voltage change less than or equal to vSmallStep:
i. The Tester shall immediately begin changing V_{BUS} voltage after the *GoodCRC* Message acknowledging the *Accept* Message. The tester shall complete the voltage transition and send *PS_RDY* before *tSrcTransSmall* expires.
ii. If the newly negotiated current is less than the previously contracted current, after *tSnkNewPower* from the EOP of *GoodCRC* acknowledging the *Accept* Message, check that the current drawn by the UUT does not exceed the newly negotiated current(I_{NEW}) . [TEST.PD.PS.SRC.3#7]
iii. If the newly negotiated current is unchanged or increased, check that the current drawn by the UUT does not exceed the previously contracted current during the transition after the EOP of GoodCRC acknowledging the *Accept* Message until the *PS_RDY* is sent by the Tester. [TEST.PD.PS.SRC.3#8]

c. If the transition is to a PPS APDO while in CV mode:
i. The Tester shall immediately begin changing V_{BUS} voltage after the *GoodCRC* Message acknowledging the *Accept* message. The Tester shall complete the voltage change and send *PS_RDY* before *tPpsSrcTransSmall* max for voltage changes less than or equal to *vPpsSmallStep*, or *tPpsSrcTransLarge* max for voltage changes greater than *vPpsSmallStep*.
ii. Check that the current drawn by the UUT does not exceed the max contracted current during and after the transition. [TEST.PD.PS.SRC.3#9]

15. If a new PDO with no change in the voltage or no new PDO is requested, and the transition involves unchanged or decreased voltage, or no new PDO is requested:
a. If the newly negotiated current is unchanged or increased: after *tSrcTransReq* max time period (before *PS_RDY* is sent) check that the current drawn from V_{BUS} does not exceed the previously contracted current. [TEST.PD.PS.SRC.3#10]
b. If the newly negotiated current is decreased: after *tSnkNewPower* from the EOP of *GoodCRC* acknowledging the *Accept* Message, check that the current drawn by the UUT does not exceed the newly negotiated current (I_{NEW}) [TEST.PD.PS.SRC.3#11]
16. The Tester sends a *PS_RDY* Message to the UUT.
17. The Tester checks that the current drawn by the UUT does not exceed the level specified in the current PDO over the next 5 seconds. [TEST.PD.PS.SRC3#12]
18. The Tester presents *SinkTxOk* if the test is in PD3 mode. The Tester waits **Error! Reference source not found.** to respond to messages from the UUT.
19. The Tester sends a new *Source_Capabilities* Message with a single PDO.
The PDO:
i. B31…30 (Fixed Supply) set to 00b.
ii. B29 (Dual-Role Power) set to 1b.
iii. B28 (USB Suspend Supported) set to 0b.
iv. B26 (USB Communications Capable) set to 0b.
v. B25 (Dual-Role Data) set to 1b.
vi. B24…22 (Reserved) set to 00b.
vii. B21…20 (Peak Current) set to 00b.
viii. B19…10 (Voltage) set to 5V.
ix. B9…0 (Maximum Current) set to the full requested current as provided in the 1st PDO of the *Sink_Capabilities* from the UUT.
20. The check fails if the UUT does not responds with a *Request* Message. [TEST.PD.PS.SRC.3#9]
21. The Tester sends an *Accept* Message.

22. If a new PDO with no change in the voltage or no new PDO is requested, and the transition involves unchanged or decreased voltage, or no new PDO is requested:
    a. If the newly negotiated current is **unchanged or increased**: after *tSrcTransReq* max time period (before *PS_RDY* is sent) check that the current drawn from V_{BUS} does not exceed the previously contracted current. [TEST.PD.PS.SRC.3#14]
    b. If the newly negotiated current is **decreased**: after *tSnkNewPower* from the EOP of *GoodCRC* acknowledging the *Accept* Message, check that the current drawn by the UUT does not exceed the newly negotiated current (I_{NEW}) [TEST.PD.PS.SRC.3#15]
23. The Tester sends a *PS_RDY* Message to the UUT.

24. The Tester checks that the current drawn by the UUT does not exceed the level specified in the current PDO over the next 5 seconds. [TEST.PD.PS.SRC.3#13]